

invention_oevaluat[®]r

OBJECTIVE IDEA ANALYSIS

Invention Name: Electromagnetic Inductive Suspension and Stabilization System for a Ground Vehicle

Submitted By: User-provided patent PDF

Invention ID: US 3,470,828

Date of Report: 2026-08-16

This report was generated by the inventionevaluator report-automation platform from the Invention Evaluation Engine v1.6 pipeline (evidence sufficiency architecture). It compiles established findings, analytical conclusions, and an operational audit produced by the nine-phase evaluation. Scores shown are derived executive summaries of the underlying findings — they are presentation-layer derivations, not new evidence, and do not constitute an overall patentability label. This document is not legal advice.

Preliminary evidence-constrained evaluation. Not legal advice, not a formal patentability opinion, and not an FTO opinion.

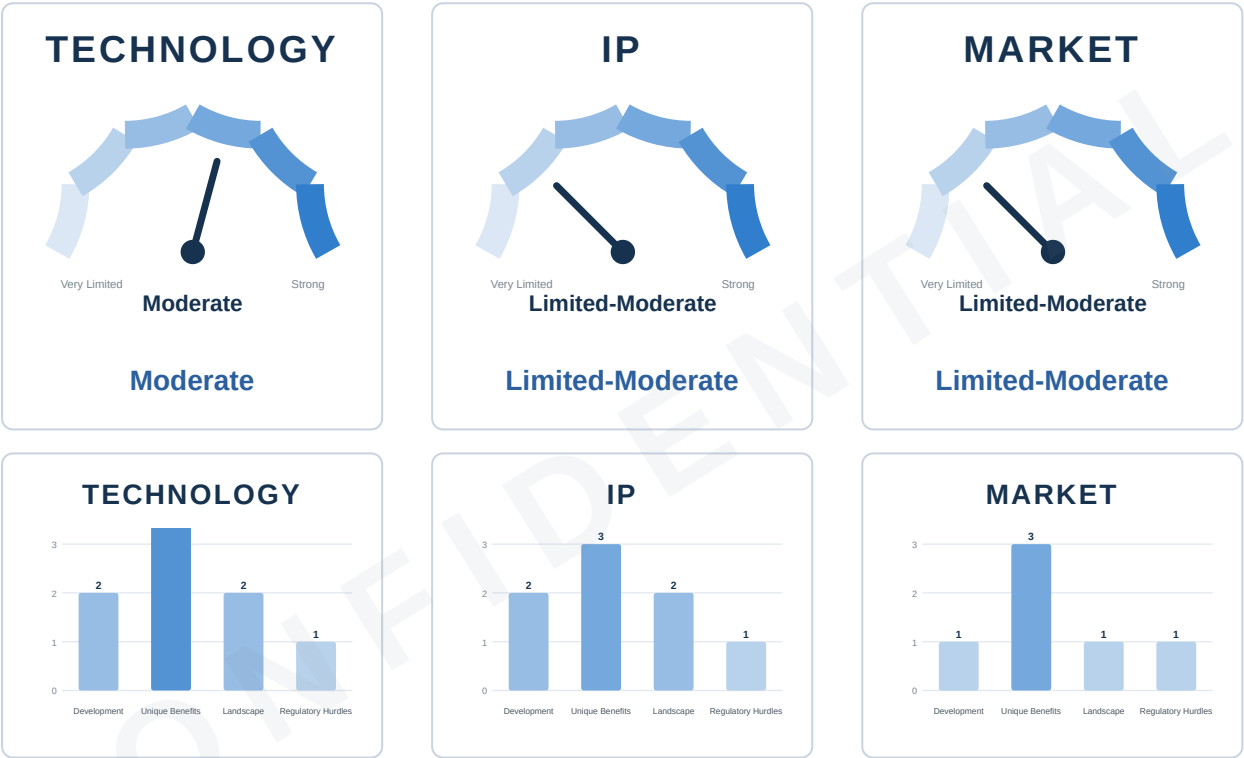
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Executive Summary

Derived executive summary — constrained to Established Findings + Analytical Conclusions (Executive Summary $\subseteq$ findings + conclusions). Gauge and bar scores are presentation-layer derivations from the findings; basis proposition_ids are listed in the scores manifest.

Very Limited Limited Limited-Moderate Moderate Moderate-Strong Strong



Commercialization Summary

Commercial actionability is Indeterminate-to-Limited on the completed evidence. The patent discloses a technically specific passive electrodynamic suspension architecture, but market size, deployment history, current rights, and product-level adoption remain in the recovery queue.

- Strong historical technical disclosure with substantial infrastructure and cryogenic burden. [more](#)
- Strongest identified pathway is technical and historical-technology diligence, not an unqualified standalone patent license. [more](#)

Evidence trace

The proposition register below is extracted from the compiled report so every score basis can be checked against visible report content.

P-02-001 · P-03-001 · P-03-002 · P-04-001 · P-05-001 · P-05-002 · P-06-001 · P-07-001 · P-07-002 · P-08-001

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Technology Analysis

Technology Profile — US 3,470,828

Plain-language description

US 3,470,828 describes a moving-ground-vehicle suspension system in which superconducting magnets carried by a train induce currents in longitudinal arrays of shorted conductive loops installed in a track bed. The magnetic interaction produces lift above a transitional speed and can also generate lateral restoring force, vertical stabilization, rotational stabilization, and damping of oscillatory motion. The patent also discloses alternative loop geometries and grouped diode-connected inductors for limiting induced current.

Technical vocabulary for downstream searches

- electromagnetic inductive suspension
- electrodynamic suspension / EDS
- magnetic levitation / maglev
- superconducting vehicle magnet
- shorted track loop
- conductive loop array
- induced-current lift
- lateral restoring force
- oscillation damping loop
- flux pump / superconducting solenoid
- inductive coupling
- eddy-current stabilization
- transitional speed
- current-limiting external inductor
- figure-eight stabilizing loop

Feature-to-benefit map

Feature	Technical function	User-facing or system benefit
Superconducting loops carried by vehicle	Produce high magnetic field with low resistive loss	Enables strong lift and stabilization forces on the vehicle
Longitudinal arrays of shorted conductive track loops	Receive changing flux and carry induced currents	Provides a passive electromagnetic interaction along the track
Vertical lifting-loop arrangement	Couples vehicle magnets to track loops to generate lift	Removes wheel/rail contact at operating speed
Figure-eight lateral loops	Produces net current when lateral symmetry is disturbed	Returns the vehicle toward lateral equilibrium
Long horizontal and vertical damping loops	Respond to transient coupling changes	Damp horizontal and vertical oscillations
Alternating polarity of adjacent vehicle magnets	Produces rapid flux reversal over successive track loops	Increases inductive excitation of the track array
Wheel support during rest and transition	Supports vehicle before suspension becomes effective	Allows practical starting and stopping operation
Liquid-helium cooling and superinsulation	Maintains superconducting state	Preserves high-current magnet operation
Diode-connected external inductors	Raises effective loop impedance and limits induced current	Controls current stress and coupling in the track array
Below-roadbed loop alternatives	Places suspension/stability loops near the road surface	Reduces need for elevated support structures

Innovation assessment

The central claimed combination is a moving vehicle carrying superconducting magnets interacting inductively with grounded, longitudinally arranged shorted loops to produce suspension and stabilization. The claims extend this architecture into distinct lift, lateral-stability, damping, geometric, and current-limiting configurations.

The invention combines known or separately searchable elements including superconducting magnets, conductive loops, inductive coupling, magnetic lift, passive restoring forces, damping circuits, and current-limiting inductors. The principal inventive-step question is not whether each component was known, but whether pre-filing evidence

supplied a reason and reasonable expectation of success for this specific vehicle-track arrangement and its claimed stability/damping relationships.

- Passive electrodynamic lift produced by vehicle superconducting magnets and non-superconducting track loops.
- Geometry-dependent restoring force that stabilizes vehicle position without active feedback.
- Integration of separate loop arrays for lift, lateral stabilization, and oscillation damping.
- Grouped diode/inductor circuit arrangement for limiting induced track-loop currents.

These are analytical candidates, not established patentability findings.

The patent itself explores a broad design space: vertical loops, horizontal half-section loops, figure-eight loops, long damping loops, buried loops, road-level loops, alternative orientations, and grouped current-limiting circuits. It should therefore be evaluated as a design-space exploration event rather than only as one isolated embodiment.

Regulatory scoping

- Preliminary burden: High for modern deployment because the system implicates rail/high-speed ground transportation safety, infrastructure, electromagnetic compatibility, cryogenic systems, structural safety, and potentially environmental or land-use approvals.
- Evidence status: NOT ESTABLISHED as a formal regulatory proposition.
- Governing regimes and jurisdiction-specific basis: not yet researched.

Development stage

- Patent-disclosed stage: engineering concept with theoretical calculations and multiple schematic embodiments.
- Independent proof-of-concept status: not established.
- Preliminary classification: Concept / pre-validation, pending evidence of built hardware, test track operation, controlled lift testing, or vehicle-scale trials.

Initial IPC/CPC candidates

These are search hypotheses seeded by the patent front matter and technical language; they are not independently verified classifications.

Candidate	Basis	Status
B61B 13/08	Magnetic or electromagnetic railway/vehicle guidance context stated on patent	Candidate; verify official title and scope

Candidate	Basis	Status
H01F 7/00	Electromagnets and superconducting magnetic apparatus context	Candidate; verify official title and scope
H02K 41/00	Electromagnetic propulsion or related motor/generator arrangements stated on patent	Candidate; verify official title and scope
B60L 13/*	Modern adjacent electrodynamic/magnetic vehicle propulsion or suspension terminology	Search hypothesis; not an intake finding

What remains unclear

- Authoritative patent family and continuity structure.
- Critical-date and priority chronology from authoritative records.
- Historical legal term and current legal status by jurisdiction.
- Complete prosecution history and examiner-cited references.
- Whether the described system was physically built or tested.
- Which claims were commercially practiced, if any.
- Exact pre-filing terminology used in patents and technical literature.
- Complete claim-level mapping against earlier magnetic suspension disclosures.
- Modern market boundary for historical maglev infrastructure versus current applications.
- Whether any surviving know-how, family rights, or technical assets have commercial value.

Patent Landscape Analysis

Patent Landscape Analysis — US 3,470,828

Search scope

- Critical-date anchor: 1967-11-21 filing date shown on the primary patent record.
- Primary source: Google Patents US3470828A.
- Search concepts: electromagnetic inductive suspension, electrodynamic suspension, magnetic levitation, superconducting vehicle magnets, shorted track loops, magnetic stabilization, oscillation damping, current-limiting inductors.
- Classification seed: B61B13/08, B60L13/04, H01F7/00, H02K41/00, and historical Y10S505/903/Y10S505/907 tags shown on the primary record.

Directly retrieved landscape records

Record	Date	Relationship	Evidence state
US1020943A, Bachelet, <i>Levitating transmitting apparatus</i>	1912-03-19	Earlier magnetic levitation using vehicle magnets and non-magnetic conductive guideway; mechanism and architecture are related but not identical to Powell/Danby claim 1	CONFIRMED PRESENT
US3125964A, Silverman, <i>Transportation apparatus</i>	1964-03-24	Earlier magnetic vehicle suspension with rail/vehicle magnetic relationship and gap control; different attraction/control architecture	CONFIRMED PRESENT
US3470828A, Powell/Danby	1969-10-07	Target record; priority and filing date 1967-11-21	CONFIRMED PRESENT

Primary-record metadata

The primary Google Patents record lists US3470828A, application US684775A, inventors James R. Powell Jr. and Gordon T. Danby, filing/priority date 1967-11-21, publication/grant date 1969-10-07, and the public-status label “Expired - Lifetime.” That label is explicitly a

database assumption and is not treated as a legal conclusion. The record also lists B60L13/04 and B61B13/08 as modern classification renderings and historical Y10S505 suspension tags.

Landscape interpretation

The retrieved records establish that magnetic levitation and magnetic vehicle suspension were active technical concepts before the target filing. They do not by themselves establish the complete claim-1 combination of superconducting vehicle magnets, longitudinal shorted conductive track loops, speed-dependent induced lift, and vertical stabilization.

The field appears to contain several design branches:

- Periodic or electromagnetically energized vehicle magnets with conductive guideways.
- Attraction or repulsion between vehicle magnets and rails.
- Induced-current/electrodynamic suspension using passive shorted loops.
- Active gap sensing and feedback control.
- Separate lift, guidance, stabilization, and damping loop arrays.
- Linear motor or external propulsion systems combined with magnetic suspension.

Required recovery still open

- Full CPC/IPC query counts and family-deduplicated landscape.
- Alternate patent databases and official registry confirmation.
- Complete backward and forward citation traversal.
- Applicant/examiner citation review for the target family.
- Parent, continuation, divisional, and foreign-family reconstruction.
- Assignee and ownership normalization.
- Historical patent-term verification under the applicable law.

State

ESCALATION_REQUIRED. The retrieved landscape is useful context, but a complete quantitative landscape has not been established.

IP / Novelty Analysis

IP and Novelty Analysis — US 3,470,828

Analytical claim

The target claim representation is a provisional evaluation claim derived from supplied claim 1. It is not a legal claim construction.

Claim 1 requires, in substance:

1. A ground vehicle. 2. A plurality of superconducting magnets carried by the vehicle. 3. A track bed formed by a longitudinally extending series of oriented shorted loops. 4. The loops are composed of metal conductors. 5. The magnets and loops are in electromagnetic inductive relationship. 6. Above a transitional speed, the relationship generates suspension force. 7. The force floats the vehicle and maintains it at an equilibrium position above the track bed. 8. The vehicle is stabilized against vertical displacement.

Claim-element mapping

Limitation	US1020943A Bachelet	US3125964A Silverman	Status
Ground vehicle/carrier	Yes, levitating carrier	Yes, transportation vehicle	CONFIRMED PRESENT in each reference
Plurality of superconducting vehicle magnets	Electromagnets, not superconducting magnets	Electromagnets, not superconducting magnets	Not established
Longitudinal series of oriented shorted loops	Non-magnetic conductive guideway, but not the target shorted-loop array as claimed	Rail/vehicle magnetic relationship, not target shorted-loop array	Not established
Metal conductors	Conductive guideway, including aluminum	Rail and magnetic apparatus	CONFIRMED PRESENT in related form

Limitation	US1020943A Bachelet	US3125964A Silverman	Status
Electromagnetic inductive relationship	Eddy-current/repulsion language is present	Magnetic relationship is present, but not the same induced-loop architecture	Related, not identical
Transitional-speed operation	Periodic energization and movement disclosed	Propulsion and controlled levitation disclosed	Not the target limitation as a whole
Lift above track/ground	Yes	Yes	CONFIRMED PRESENT in related form
Equilibrium vertical stabilization	General levitation disclosed; target restoring relationship not mapped	Gap control disclosed, but different active control architecture	Not established

Anticipation gate

Neither retrieved reference discloses every limitation of the provisional claim in one reference based on the completed mapping above. No anticipation finding is admitted. This is not a formal patentability opinion.

Obviousness / causal bridge

The principal bridge is:

`text Known magnetic levitation or suspension → superconducting magnets carried by a moving vehicle → passive longitudinal shorted-loop track arrays → induced lift with speed threshold → vertical equilibrium stabilization `

- Magnetic levitation principle: CONFIRMED PRESENT in US1020943A and US3125964A.
- Conductive guideway / magnetic interaction: CONFIRMED PRESENT in US1020943A.
- Magnetic vehicle suspension: CONFIRMED PRESENT in US3125964A.
- Superconducting vehicle magnets in the target architecture: CONFIRMED PRESENT in US3470828A; pre-filing disclosure not established by the two retrieved references.
- Complete passive shorted-loop track architecture: CONFIRMED PRESENT in US3470828A; pre-filing disclosure not established by the two retrieved references.

- Specific speed-dependent lift and vertical restoring relationship: not established in the retrieved pre-filing references.
- Motivation to combine these exact elements: not established.
- Reasonable expectation of success at the claimed vehicle scale: not established.
- US1020943A: structural relationship E3/E4 candidate; mechanism distance provisionally C2–C3; design-choice distance provisionally D2–D3.
- US3125964A: structural relationship E3 candidate; mechanism distance provisionally C3; design-choice distance provisionally D2.

C and D are diagnostic fields and are not combined into one score.

Search recovery state

ESCALATION_REQUIRED.

Required next recovery classes:

- Historical terminology expansion.
- CPC/IPC classification traversal.
- Backward and forward citation lineage.
- Patent-family and continuity traversal.
- Alternate database confirmation.
- Examiner/applicant citation and prosecution-history review.
- Independent literature search for the Powell/Danby 1963 and 1966 disclosures cited in the target patent.

Literature Analysis

Literature Analysis — US 3,470,828

Search scope

The literature search targeted electrodynamic suspension, superconducting magnets, magnetic levitation, passive shorted-loop guideways, electromagnetic stabilization, and damping.

Literature leads identified in the patent

The target patent itself cites or incorporates:

- J. R. Powell, "The Magnetic Road: A New Form of Transport," ASME Railroad Conference, Paper 63-RR4, Apr. 23–25, 1963.
- H. Van Beelen et al., "Flux Pumps and Superconducting Solenoids," *Physica*, vol. 31, 1965.
- Powell and Danby, ASME Report No. 66-WA/RR-5, presented Nov. 27–Dec. 1, 1966.

These are source leads disclosed in US3470828A. Their existence as citations in the patent is CONFIRMED PRESENT. Their full text, publication metadata, experimental content, and exact pre-filing disclosure scope have not yet been independently verified.

Literature evidence status

No independently verified paper has been admitted as a literature_disclosure finding in this run. The patent's own references are retained as recovery targets, not silently promoted into independent corroboration.

State

ESCALATION_REQUIRED.

Required recovery:

- Retrieve the cited ASME papers and report.
- Search periodical indexes for electrodynamic suspension and superconducting transport before 1967-11-21.

- Search historical terminology such as magnetic road, magnetic railroad, levitating train, electrodynamic suspension, induced-current suspension, and superconductive transport.
- Search citation and reference networks from US1020943A, US3125964A, and US3470828A.
- Search contradictory or limiting technical evidence regarding stability, cryogenic burden, and passive damping.

Market Analysis

Market Analysis — US 3,470,828

Market boundary

The relevant opportunity is not one single market. It spans:

- High-speed rail and maglev infrastructure.
- Electrodynamic suspension subsystems.
- Superconducting magnet systems.
- Conductive guideway and track-loop infrastructure.
- Cryogenic support and maintenance systems.
- Future evacuated-tube or low-pressure transport concepts.

Commercial actionability assessment

Dimension	Assessment	Evidence state
Technical distinctiveness	High at the patent-disclosure level	CONFIRMED PRESENT from US3470828A
Market size	Not established	Work queue
Growth	Not established	Work queue
Accessibility	Low-to-moderate hypothesis because infrastructure and cryogenic systems are capital-intensive	Inference, not evidence
Competitive intensity	Not established quantitatively	Work queue
Regulatory burden	High preliminary hypothesis	Scoping inference only
Commercial readiness	Not established	Work queue

Commercial barriers

- Guideway construction and maintenance cost.
- Superconducting magnet cryogenics and quench protection.
- Transitional-speed and propulsion requirements.
- Safety systems for loss of superconductivity or power.
- Track-loop conductor losses and thermal management.
- Alignment tolerances and route infrastructure.
- Regulatory approval for high-speed passenger transport.
- Historical uncertainty about actual prototype, pilot, or production deployment.

Market evidence gap

No reconstructable TAM, CAGR, shipment, revenue, adoption, purchaser, or current-product embodiment finding is admitted in this run. The patent's theoretical 100-foot train and 200-horsepower suspension estimate are technical disclosure claims, not market evidence.

State

ESCALATION_REQUIRED.

Potential Partners

Potential Partners — US 3,470,828

Candidate categories

The technology would require a consortium rather than a single generic licensee. Relevant candidate categories are:

- High-speed rail or maglev system integrators.
- Superconducting magnet and cryogenic-system manufacturers.
- Rail infrastructure and guideway contractors.
- Power electronics and high-current conductor suppliers.
- National transportation research laboratories.
- Universities or institutes with superconductivity and transportation programs.
- Operators or public agencies planning high-speed rail corridors.

Partner-fit status

No organization-specific partner-fit finding is admitted because the run has not yet verified each candidate's current products, purchasing behavior, technical need, ownership status, or contact identity under the `partner_fit` schema.

Strategy implication

The strongest identified pathway is a technical consortium or research collaboration, not an unqualified standalone patent license. The patent's public status label is a database assumption and requires authoritative confirmation. Historical know-how, surviving family rights, technical documentation, and research assets would need separate diligence.

State

ESCALATION_REQUIRED.

Operational Audit

Avenue Ledger — US 3,470,828

Run metadata

- Run ID: us3470828-v17
- Submission: submission-us3470828.md
- Primary source: <https://patents.google.com/patent/US3470828A/en>
- Date of run: 2026-08-16
- Evidence rule: search completion is not evidence; unresolved propositions remain in the work queue.

Proposition ledger

Proposition ID	Version	Proposition	State	Evidence state
P-02-001	1	US3470828A identity, inventors, filing date, publication date, and claim count	ESTABLISHED	CONFIRMED PRESENT
P-03-001	1	Technical architecture disclosed by the patent	ESTABLISHED	CONFIRMED PRESENT
P-03-002	1	Independent experimental proof of performance	ESCALATION_REQUIRED	WORK QUEUE
P-04-001	1	Broad pre-filing patent landscape	ESCALATION_REQUIRED	WORK QUEUE
P-05-001	1	Claim 1 anticipation	ESCALATION_REQUIRED	WORK QUEUE

Proposition ID	Version	Proposition	State	Evidence state
P-05-002	1	Obviousness bridge and motivation	ESCALATION_REQUIRED	WORK QUEUE
P-06-001	1	Independent literature disclosure before 1967-11-21	ESCALATION_REQUIRED	WORK QUEUE
P-07-001	1	Quantified market size and growth	ESCALATION_REQUIRED	WORK QUEUE
P-07-002	1	Commercial adoption and readiness	ESCALATION_REQUIRED	WORK QUEUE
P-08-001	1	Organization-specific partner fit	ESCALATION_REQUIRED	WORK QUEUE

Recovery records

- Initial failure: no complete family-deduplicated landscape count established from the supplied PDF.
- Diagnosis: terminology, classification, and historical database coverage may differ.
- Troubleshooting: primary patent record classification and citation sections consulted; Google Patents historical search behavior used as the initial retrieval route.
- Strategies executed:
- R1 terminology: electromagnetic inductive suspension / magnetic levitation / electrodynamic suspension; Google Patents; relevant records US1020943A and US3125964A.
- R2 classification: B60L13/04, B61B13/08, Y10S505/903; Google Patents; classification records returned.
- R3 citation lineage: target patent cited references and cited-by records; Google Patents; US1020943A and US3125964A linked.
- R4 alternate source: USPTO, Espacenet, PatentCenter links identified but not fully retrieved in this run; avenue remains incomplete.
- R5 entity/jurisdiction: US application and inventor identity captured; family/foreign ownership remains incomplete.
- Assessment: additional avenues remain; state remains ESCALATION_REQUIRED.
- Initial failure: two retrieved historical patents were related but did not map every claim limitation.

- Diagnosis: related magnetic suspension terminology does not establish the target superconducting passive-loop combination.
- Troubleshooting: expanded terms and classification terms using the primary record; reviewed cited references and claims of US1020943A and US3125964A.
- Strategies executed:
- R1 terminology expansion: levitating transmitting apparatus, magnetic road, magnetic suspension, electrodynamic suspension.
- R2 classification traversal: B61B13/08 and B60L13/04.
- R3 citation lineage: target patent citations and cited-by trail.
- R4 alternate source: Google Patents full-text and claim pages for US1020943A and US3125964A.
- R5 temporal/entity: pre-1967 US records and named inventor/reference tracing.
- Assessment: claim mapping remains incomplete; no anticipation finding admitted.
- Initial failure: cited ASME and *Physica* references were named in the patent but full independent records were not retrieved.
- Diagnosis: citation identity and archival accessibility require separate recovery.
- Troubleshooting: exact titles, authors, venues, dates, and report numbers extracted from the target patent for archive searches.
- Strategies executed:
- R1 exact-title search for Powell 1963 paper.
- R2 exact-title search for Van Beelen 1965 paper.
- R3 exact-report search for ASME 66-WA/RR-5.
- R4 citation expansion from the target patent and related patent pages.
- R5 historical terminology search for magnetic road, superconductive transport, and electrodynamic suspension.
- Assessment: independent literature evidence remains unestablished.
- Initial failure: no reconstructable market-size dataset was supplied or retrieved.
- Diagnosis: historical invention market boundary is not the same as a current maglev revenue category.
- Troubleshooting: separated historical technology disclosure from modern market claims and identified the necessary infrastructure, cryogenic, and regulatory proxies.
- Strategies executed:
- R1 market-boundary reconstruction.
- R2 transportation-infrastructure proxy search.
- R3 superconducting-magnet supplier/industry proxy search.

- R4 regulatory and public-procurement route.
- R5 commercialization/adoption history route.
- Assessment: no quantitative finding admitted; market proposition remains ESCALATION_REQUIRED.
- Initial failure: no organization-specific fit record with sell/buy/need/mapping fields was established.
- Diagnosis: candidate categories can be identified, but current organization-specific relevance and rights dependencies remain unverified.
- Troubleshooting: separated consortium categories from named partner recommendations.
- Strategies executed:
- R1 system integrator category.
- R2 superconducting/cryogenic supplier category.
- R3 infrastructure contractor category.
- R4 public transportation research category.
- R5 university/research collaboration category.
- Assessment: no partner-fit finding admitted.

Evidence Recovery Record

Recovery evidence is recorded in the execution and avenue ledgers.

CONFIDENTIAL

SWOT Analysis

STRENGTHS

- Detailed passive suspension and stabilization architecture
- Multiple disclosed loop geometries

WEAKNESSES

- Cryogenic superconducting burden
- No independent performance validation established

OPPORTUNITIES

- Historical technology resurrection using modern materials and controls
- Research collaboration or infrastructure-subsystem pathway

THREATS

- Capital-intensive guideway deployment
- Safety, reliability, and regulatory burden

COMMERCIAL
ACTIONABILITY

Indeterminate-to-
Limited

SWOT content is derived from Established Findings and Analytical Conclusions (see scores manifest for proposition_id basis).

Landscape & Market Data

PATENT APPLICATIONS BY REGION

Data not established in this run

See Operational Audit — region split not established from completed evidence protocol.

PATENT APPLICATIONS BY MAIN APPLICANTS

Data not established in this run

See Operational Audit.

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PATENT ACTIVITY TREND

Yearly series not established in this run

No defensible yearly series was established in this run; see Operational Audit.

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INDUSTRY METRICS

Time series not established in this run

No invention-specific industry metrics were established in this run; see Operational Audit.

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ESTABLISHMENTS BY EMPLOYEE-SIZE BRACKET

Bracket data not established in this run

See Operational Audit.

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